

Highly sensitive formaldehyde sensors based on polyvinylamine modified polyacrylonitrile nanofibers†

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A novel, sensitive, selective, and reproductive formaldehyde sensor has been developed by coating polyvinylamine (PVAm) modified electrospun polyacrylonitrile (PAN) nanofibrous membranes on quartz crystal microbalance (QCM) for the first time. Benefiting from the abundant primary amine groups in PVAm, large specific surface area, high porosity and hierarchical structure of PAN nanofibers, and the strong interaction between PVAm molecule and formaldehyde, the as-prepared QCM sensors achieved an extremely low detection limit of 500 ppb in a remarkably short period of time (120 s). Moreover, the sensitivity of the fibrous membranes based sensors are 2.5 times higher than that of flat film-based ones when upon exposure to 500 ppb formaldehyde. Furthermore, the as-prepared QCM sensors possess excellent reproducibility, reversibility, and good selectivity by virtue of the reversible nucleophilic addition reaction between formaldehyde molecule and primary amine group in PVAm. Hence, such promising QCM sensors could not only potentially allow for monitoring gaseous formaldehyde, but also pave a way for designing and development of novel QCM sensing systems based on functionalized nanofibrous membranes.

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Introduction

Formaldehyde is one of the most abundant air pollutants which could cause a great deal of diseases such as childhood asthma, sick building syndrome, or even cancer.^{1–4} To date, numerous approaches such as high performance liquid chromatography (HPLC), electrochemical methods, quartz crystal microbalance (QCM) sensors have been developed for the detection of formaldehyde.^{5–7} Among them, QCM sensors have attracted tremendous attention due to its accurate, real-time detection, and easy to use.

QCM is an acoustic wave technique based on the piezoelectric effect, which has been widely used for accurate and real-time monitoring tiny mass changes of even less than nanogram.^{8–10} According to this mechanism, QCM-based sensors could be utilized to detect materials in extremely low concentration levels by coating sensitive materials which can adsorb analytes efficiently on QCM electrode. For example, molecular imprinted polymer (MIP) could be coated onto the

electrode of the QCM as a thin permeable film for the detection of formaldehyde, benzene, terpenes, and nandrolone.^{11–14} However, the poor surface area of the flat films have restricted the as-prepared sensors for highly sensitive determination of formaldehyde. Hence, it encouraged researchers to deeply study the factors which influence the sensing performance. Significantly, besides the structures of the instruments, the properties of sensing layers are the most important factors constrained the performance of a QCM-based sensor: firstly, the adsorption sites of the sensing layer would be dramatically increased by enlarging the surface area of sensing materials, and subsequently leading to a remarkable increase on adsorption amount and sensitivity potentially;^{15,16} secondly, the energy consumption during vibration transfer would be decreased if the rigidity of the coating layer is enhanced, and consequently promoted the sensitivity and response speed of the sensors;^{17,18} finally, the porous structure of the coating layer would accelerate the adsorption and desorption process potentially, thus improve the response rate.¹⁹ With these in mind, considerable research efforts have been devoted to develop novel QCM sensors which possess excellent properties by combining QCM technique with other technologies to regulate the coating layers.

Electrospinning has been established as an attractive approach for manufacturing nanofibers with controllable compositions and structures.^{20–24} The nanofibers with large surface area, high porosity, and large stacking density would be an ideal candidate to replace the flat films to enhance the QCM-based sensors' performances.^{25–27} For instance, Wang *et al.*²⁸

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prepared a QCM sensor for determination of formaldehyde by spun polyethyleneimine (PEI) and polyvinyl alcohol (PVA) composite nanofibers on QCM electrode, the as-prepared sensors achieved a much lower detection limit (10 ppm) than PEI-flat film based ones. However, the hygroscopicity of PVA nanofibers made the sensors affected by the humidity easily, and the content of primary amine groups was not enough to increase the sensitivity dramatically (the primary amine groups could take place nucleophilic addition reaction with formaldehyde molecule). Hence, it is emerged to explore new polymer template and sensing materials to improve the sensitivity of QCM sensors.

On the other hand, polyvinylamine (PVAm) is a newly developed class of cationic polymer with a wide range of interesting features including abundant activated primary amine groups, high charge density, and controllable molecular chains length which make it a promising compound to be utilized as a sensing material to detect formaldehyde.²⁹ Polyacrylonitrile (PAN) molecule possess a relatively large rigidity due to the intensive polarity and strong dipole force of cyano groups, meanwhile, PAN nanofibers own large surface area, high porosity, hydrophilicity, and low hygroscopicity, these properties make the as-spun nanofibers an ideal candidate to be utilized as an appropriate coating material for QCM sensors.^{30–32} To the best of our knowledge, no effort has been given to the development of PVAm modified PAN nanofibers coated QCM sensors.

In this contribution, we presented the fabrication of a QCM-based sensor by coating PVAm modified electrospun PAN nanofibrous membranes (PAN–PVAm) on the surface of electrode for the detection of formaldehyde for the first time. (Fig. 1) The sensing properties of the as-prepared of as prepared QCM sensors have been systematically investigated at room temperature. Taking advantage of the large surface area, high porosity, and high rigidity of PAN fibrous membranes and the abundant primary amine groups of PVAm molecule (in order to make comparison, the molecular geometry of PEI is shown in Fig. S1†), the as-prepared sensors exhibit significantly enhanced sensing performance, such as rapid response, good selectivity, excellent reproducibility, and high sensitivity, thus showing a promising application for routine formaldehyde sensors.

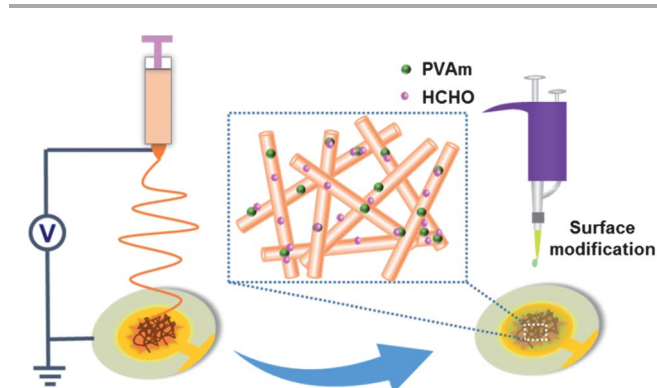


Fig. 1 Schematic illustration for the preparation of sensing PAN–PVAm NFM coated on a QCM electrode.

Experiment

Materials

PAN ($M_w = 90\,000$) was purchased from Spectrum Chemicals & Laboratory Products Co., Ltd., USA. PVAm (23 wt%, $M_w = 100\,000$) was supplied from Mitsubishi Chemical Holdings Corporation. Formaldehyde (GCS grade) was brought from Aladdin Chemical Reagent Co., Ltd., China. *N,N*-dimethylformamide (DMF), chloroform, ethanol, acetone, and other volatile organic compounds (VOCs, analytical grade) were obtained from Shanghai Chemical Reagents Co., Ltd., China. Ultrapure water with a resistance of 18.2 MΩ was prepared by using a Heal-Force system. All chemicals were used as received without further purification.

Preparation of fibrous PAN–PVAm membranes on QCM electrode

The 6 wt% PAN solution was prepared by dissolving the PAN powder into DMF with vigorous stirring for 12 h. A typical procedure (Fig. 1) for fabricating PAN nanofibers on QCM electrode was performed by using DXES-1 spinning equipment (Shanghai Oriental Flying Nanotechnology Co., Ltd., China) with an applied voltage of 25 kV and a flow rate of 1 mL h^{−1}. The fabrication chamber was kept at constant temperature (25 °C) and relatively humidity (45%). The PAN fibrous membranes were deposited on the grounded QCM electrode (5 MHz, AT-cut quartz crystal with Au electrodes) at a 15 cm tip to collector distance until the required frequency shift was obtained. The resonance frequencies were measured by a QCM digital controller (QCM1000, Nanosensing Research Systems, China). The fibrous membranes coated QCMs were dried at 70 °C in a vacuum for 1 h prior to the subsequent characterization. PVAm (1 wt%) aqueous solution was dropping casted onto fibrous PAN membranes by a micropipettor (5–50 μL, Shanghai Liansheng Instrumental Factory, China), through which fibrous PVAm functionalized PAN membranes coated QCM sensor was prepared. After the solvents were completely evaporated at room temperature, the membranes coated QCM sensors were dried at 40 °C for 2 h in vacuum. The morphologies of the obtained fibrous membranes were examined by field emission scanning electron microscopy (FE-SEM) (S-4800, Hitachi Ltd., Japan).

Formaldehyde sensors configuration and experiment setup

The resonance frequency of a QCM is related to the mass change due to the adsorption of analytes onto the surface of the sensing materials, which can be calculated by Sauerbrey equation as follows,³³

$$\Delta f = \frac{f_0^2 \Delta m}{A(\mu\rho)^{1/2}} \quad (1)$$

where Δf is the frequency shift (Hz), f_0 is the fundamental frequency of a bare QCM chip, Δm is the mass change per unit area (g cm^{−2}), A the surface area of the electrode (1.00 cm²), ρ is the density of quartz (2.648 g cm^{−3}), and μ is the shear modulus of quartz crystal (2.947×10^{11} dyne cm²). As a result, the

frequency is decreased by 1 Hz in the case of 16.67 ng of mass being loaded for a 5 MHz QCM.

A schematic diagram of a static-type formaldehyde testing system is shown in Fig. S2.† The sensor was installed in the testing chamber (7.64 L) which was kept at constant temperature of 25 °C and relative humidity of 60%. A microliter syringe (1–50 µL, Shanghai Anting Microliter Syringe Co., China) was used for the analytes injections. The concentration of the injected analyte in the chamber was calculated by the following formula,³⁴

$$C = \left(\frac{22.4w\rho_1 V_s T}{273MV} \right) \times 10^3 \quad (2)$$

where C is the analyte concentration in ppm, w is the liquid analyte mass concentration in wt%, ρ_1 is the density of liquid analyte in g mL^{-1} , V_s is the volume of the analyte in μL , T is the temperature in testing chamber in Kelvin, M is the molecular weight of analyte in g mol^{-1} , and V is the chamber volume in L. The sensing properties of the sensors were examined by measuring the resonance frequency shift of the QCM aroused by the additional mass loading. The data from the sensors were recorded by a personal computer. In addition, to desorb the testing vapor from the sensing membranes, a flow of high-purity N_2 was purged through the chamber.

Results and discussion

Morphology and structure of nanofibrous membranes

The representative FE-SEM images of the PAN fibrous membranes and PVAm functionalized PAN membranes were carefully studied, as shown in Fig. 2. The as-spun PAN nanofibers were randomly oriented three-dimensional (3D) porous structures with a relatively uniform diameter of 136 nm and a standard deviation is 20 nm. (Fig. 2a and b) After modified with PVAm molecule, the 3D porous structure of the fibrous membranes was still maintained without obvious change, but the diameters of the PAN–PVAm nanofibers were changed to 225 nm with a standard deviation of 28 nm. It is worth noting that amounts of conglutinations was observed among adjacent fibers, (Fig. 2c) this phenomenon may be caused by the strong

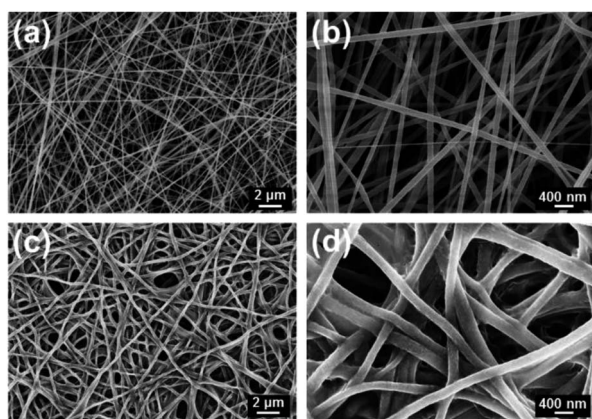


Fig. 2 FE-SEM images of (a and b) PAN nanofibrous membranes and (c and d) PVAm modified PAN nanofibrous membranes.

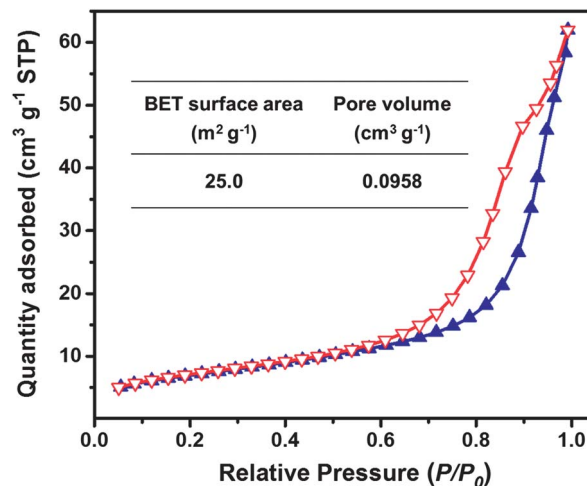


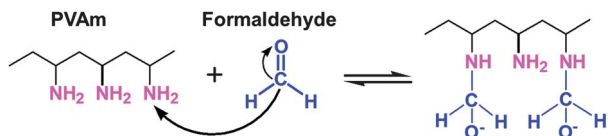
Fig. 3 N_2 adsorption-desorption isotherms of PAN membranes. The inset shows the BET surface area and pore volume, respectively.

hydrogen bond between cyano group ($\text{C}\equiv\text{N}$) in PAN and abundant primary amine groups ($-\text{NH}_2$) in PVAm.^{35,36} Moreover, this interesting structure would further enhance the interaction between sensing layer and QCM electrode, thus improve the stability of the sensing layer.³⁷ Additionally, deep sighting in the FE-SEM images displayed a relatively high surface roughness on fibers by forming some protuberances which could be due to the irregular deposition of PVAm membrane.^{38,39} (Fig. 2d) This fantastic structure could potentially remarkably increase the surface area, and it also drove us to analyse the pore structure of the fibrous membranes carefully. The Brunauer-Emmett-Teller (BET) surface area and pore structure of PAN membranes were examined by the N_2 adsorption/desorption test with a surface area analyzer (ASAP 2020, Micromeritics Co., USA). The N_2 adsorption-desorption isotherms displayed in Fig. 3 exhibited a type IV with a H3 hysteresis loop, suggesting the characteristics of mesopores within the as-prepared membranes.^{40,41} Taking advantage of the small diameter and porous structure, the as-spun PAN nanofibrous membranes, possessed a BET surface area of $25.0 \text{ m}^2 \text{g}^{-1}$ and a pore volume of $0.094 \text{ cm}^3 \text{g}^{-1}$, respectively.

Accordingly, such PAN fibrous membranes with large surface area, high porosity and hierarchical structure which could provide abundant adsorption sites are appropriate to act as formaldehyde QCM sensors' template.

Formaldehyde gas sensing characteristics

The formaldehyde sensing properties of QCM-based sensors coated with PAN–PVAm fibrous membranes were investigated by measuring the frequency shift when the as-prepared sensors were exposed to formaldehyde gas. Due to the reversible nucleophilic addition interaction between formaldehyde molecule and primary amine groups of PVAm,^{42,43} (Scheme 1) the formaldehyde molecule would be adsorbed onto the surface of sensing layer and led to the response amplitude increase. As expected, on each injection of formaldehyde into the chamber, the response underwent a dramatic drop and reached a steady



Scheme 1 Illustration of the nucleophilic addition between formaldehyde and PVAm.

value after approximately 120 s (Fig. 4a). In addition, the frequency shifts were gradually increased with formaldehyde concentration increasing. The maximum frequency shift of PAN-PVAm flat film coated QCM based QCM sensors upon exposure to 0.5, 1, 5, 15, 35 ppm of formaldehyde were 0.2, 0.4, 0.7, 2.5, 4.2 Hz, respectively. In comparison, the PVAm modified PAN fibrous membranes coated QCM sensors showed a remarkable frequency shift (0.5 Hz) to 500 ppb of formaldehyde, which is 2.5 times higher than that of flat film based sensors. This further promoted sensitivity could be attribute to the large surface area, high porosity of the PAN nanofibers. More importantly, the QCM-based PAN-PVAm NFM sensors exhibited a reduced detection limit for formaldehyde (500 ppb)

compared with those reported in literatures, which demonstrated that the QCM sensors coated with PEI, MIP, and calixarene flat films have detection limit of 1 ppm,³⁴ 28 ppm,⁴⁴ and 109 ppm (ref. 45) for formaldehyde. Moreover, the sensitivity change of the QCM sensors with PAN-PVAm flat film and fibrous membranes coatings as a function of formaldehyde concentration in the range of 0.5 to 35 ppm was plotted in Fig. 4b. It suggested that the as-prepared sensors presented a good linear dependency between the response and formaldehyde concentration. In addition, it is worth noting that the adsorption of the formaldehyde on sensing layer showed a Henry-type sorption⁴⁶ at the concentration range, which is in good agreement with the result reported in our previous work.³⁹ Considering the safe-exposure limit set by the Occupational Safety and Health Administration standard (OSHA, 750 ppb),⁴⁷ the as-prepared PAN-PVAm NFM coated QCM sensors with low detection limit (500 ppb) and fast response (approximately 120 s) could potentially be used for routine detection of formaldehyde.

In order to clarify the influence of PAN nanofibers depositing quantity on sensors' sensitivity, the performance of the sensors with different PAN nanofibers coating loads were investigated by fixing the coating loads of PVAm (3500 Hz). It could be found that the QCM based PAN-PVAm NFM sensors with 1000 Hz coating load of PAN nanofibers exhibited the highest response amplitude towards each concentration (7.9 to 35 ppm of formaldehyde), further increasing the coating load (2000 Hz) would decrease the frequency response (Fig. 5). This phenomenon might be explained by the following reasons: firstly, the softness of the nanofibrous membranes led to the energy dissipation during vibration transfer;^{17,48} secondly, the PVAm molecule may access into the inner of the PAN NFM when depositing PVAm on the surface of the membranes, which would prevent the adequate contacting between PVAm adsorption sites and analytes.³⁶ In this case, the PAN nanofibers' coating load was fixed at 1000 Hz to ensure the accuracy of the experiment.

As such, a series of QCM sensors coated with varied casing loads of PVAm were prepared to investigate the relationship of

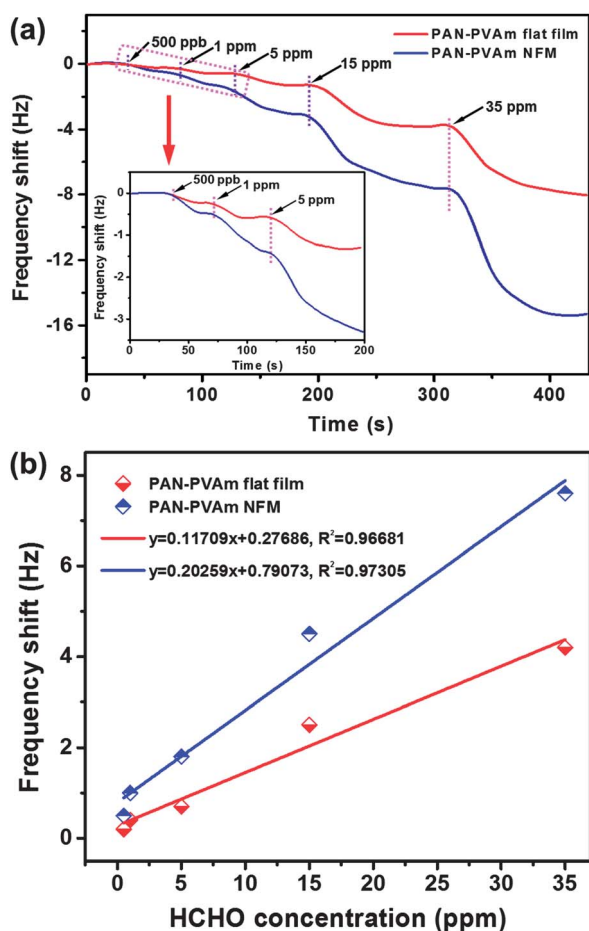


Fig. 4 Dynamic response of QCM sensors coated with two different sensing structures upon exposure to increasing formaldehyde concentrations. Inset is the amplified image for 0.5, 1, and 5 ppm formaldehyde detection. (b) The frequency shift versus formaldehyde concentration. The coating loads of PAN nanofibers and PVAm were 1000 and 3500 Hz.

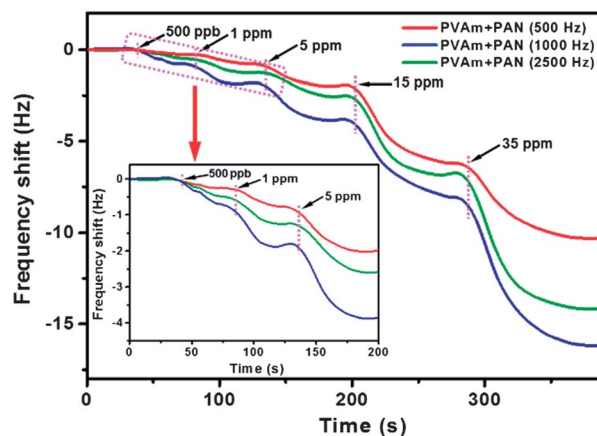


Fig. 5 Dependence of frequency shift for QCM-based PAN-PVAm NFM sensors with various PAN coating loads on formaldehyde (0.5–35 ppm). The coating loads of PVAm were 3500 Hz.

frequency shifts and coating loads of sensing materials, as illustrated in Fig. 6. The response amplitudes towards each concentration of formaldehyde were increased with the coating load of the PVAm increasing because of the much more adsorption sites brought about by the increasing of PVAm. When the QCM-based PAN-PVAm NFM sensors with various PVAm coating loads (1500, 3500, and 5500 Hz) were exposed to 35 ppm of formaldehyde, the maximum frequency shift were up to 1.1, 7.4, and 10.0 Hz, respectively. The calibration curve shown in Fig. 6b further confirms that the response of QCM sensors upon exposure to 35 ppm formaldehyde was increased with the amount of PVAm deposited on the electrode. It should be noted that the response reached a steady value at a higher PVAm coating load of 5500 Hz. Although increasing the PVAm coating load might increasing the adsorption of formaldehyde, more PVAm deposited on the PAN nanofibers could lead to more obvious conglutination structures which may decrease the surface area of the sensing membranes and thus lower the sensors performance. This results were consistent with our previous study that a large amount of PEI deposited on the fibrous membranes decreased the surface area of sensing layers, which played a negative role in improving the sensitivity

of resultant gas sensors.³⁸ On the basis of the above discussion, we could reasonably conclude that PVAm molecules played a critical role in sensing of formaldehyde, and highly precise manipulation of PVAm coating load should be adopted to guarantee the accuracy and comparability of detection results.

With regard to practical use, the selectivity of the resultant sensors is investigated by exposing the sensors to a variety of possible interference VOCs at the concentration of 15 ppm, including DMF, acetone, ethanol, benzene, toluene, chloroform, and tetrahydrofuran (THF). It is obvious that the as-prepared QCM sensors based on PVAm modified PAN nanofibers showed the highest response to formaldehyde with a value up to 4.2 Hz (Fig. 7), but slight frequency shift to other interference VOCs (less than 0.5 Hz) (inset of Fig. 7). The diversity of the response indicated that the adsorption induced by the nucleophilic addition between formaldehyde and PVAm is the main reason to cause the additional mass loading, and the physisorption on the surface of the membranes and gold electrode is also existed. The highly selectivity to formaldehyde makes the sensors excellent candidates for practical gas sensing application.

Reproducibility is considered to be one of the most significant factors that affect the eventual applicability of chemical sensors. To examine the reproducibility of the resultant sensors, after reaching the equilibrium, a flow of high-purity N₂ was purged through the chamber to desorb the analytes adsorbed on the surface of the membranes, and then testing the formaldehyde sensing properties under the same conditions. It could be seen that the sensors showed nearly the same frequency shift upon exposure to each concentration at different two cycles, and the equilibrium frequency of the sensors could shift to its initial value after desorption (Fig. 8). The results implied that the PAN-PVAm NFM based QCM sensors possess excellent reproducibility attributed to the reversal nucleophilic addition between PVAm and formaldehyde, and could be potentially used to monitor formaldehyde in indoor air.

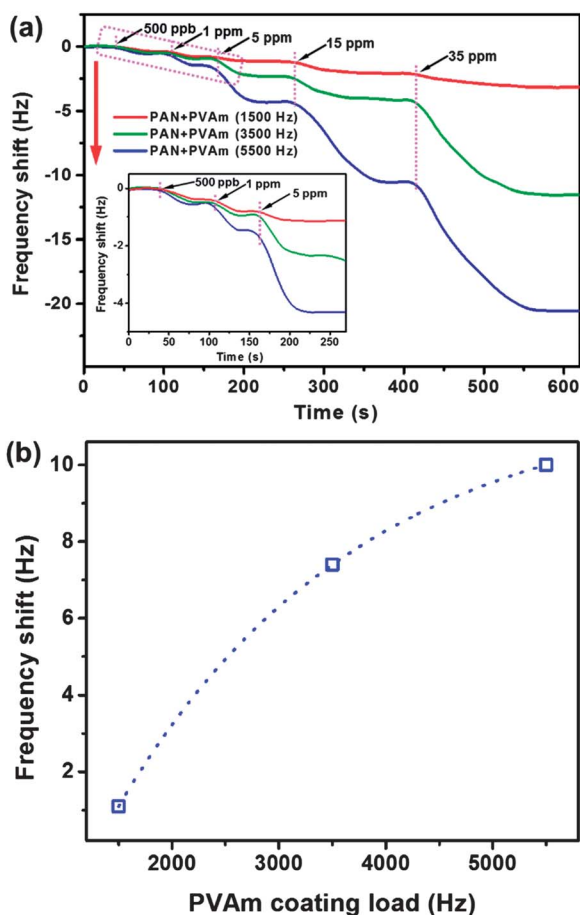


Fig. 6 (a) Response of QCM-based PAN-PVAm NFM sensors with various PVAm coating loads upon exposure to different concentration formaldehyde (0.5–35 ppm). (b) Frequency shift versus PVAm coating load upon exposure to 35 ppm formaldehyde. The coating loads of PAN nanofibers were 1000 Hz.

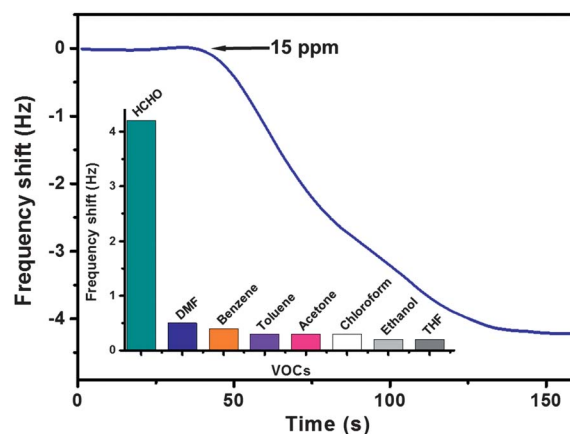


Fig. 7 Response of QCM-based PAN-PVAm NFM sensors upon exposure to 15 ppm of formaldehyde. The insets are the frequency response of QCM-based sensors versus 15 ppm of various VOCs. The coating loads of PAN and PVAm were 1000 and 3500 Hz, respectively.

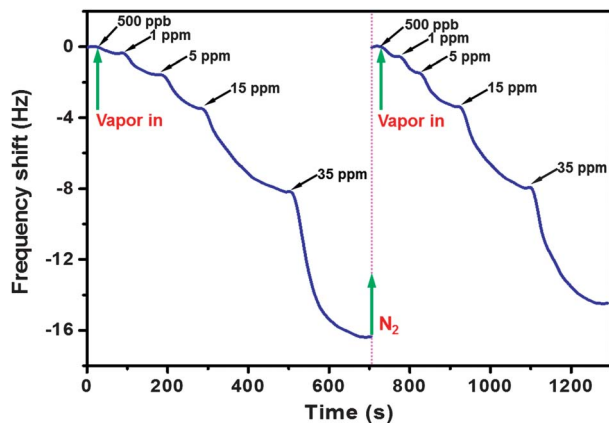


Fig. 8 Reproducibility test for QCM-based PAN-PVAm NFM sensors upon exposure to formaldehyde. The coating loads of PAN and PVAm were 1000 and 3500 Hz, respectively.

Conclusions

In summary, we have demonstrated the first attempt to apply PAN-PVAm NFM as sensing layers on QCM to construct formaldehyde sensors. The large surface area, high porosity, and large stacking density of PAN fibrous membranes provide extensive active sites for formaldehyde adsorption, and consequently achieved an extremely low detection limit of 500 ppb which is much lower than OSHA's standard. Additionally, the sensitivity of fibrous membranes coated QCM sensors are 2.5 times higher than that of flat-film based ones. Moreover, the influence of PAN nanofibers coating loads and PVAm casting loads were investigated carefully. Furthermore, the as-prepared QCM sensors possess excellent reproducibility and good selectivity due to the reversible nucleophilic addition reaction between formaldehyde molecules and primary amine groups in PVAm. The excellent sensing performance implied that the PAN-PVAm NFM coated QCM sensors will be potentially used for the routine determination of formaldehyde. Moving forward, this work also promotes a strategy for the design and development of versatile QCM sensors toward various analytes in the future.

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Notes and references

- 1 G. D. Nielsen and P. Wolkoff, *Arch. Toxicol.*, 2010, **84**, 446.
- 2 R. Chenier, *Hum. Ecol. Risk Assess.*, 2003, **9**, 509.
- 3 J. D. Thrasher and K. H. Kilburn, *Arch. Environ. Occup. Health*, 2001, **56**, 311.
- 4 K. Xiao, J. Luo, W. L. Fowler, Y. Li, J. S. Lee, L. Xing, R. H. Cheng, L. Wang and K. S. Lam, *Biomaterials*, 2009, **30**, 6016.

- 5 M. Hammerle, E. A. H. Hall, N. Cade and D. Hodgins, *Biosens. Bioelectron.*, 1996, **11**, 246.
- 6 P. W. Wu, C. C. Chang and S. S. Chou, *J. Food Drug Anal.*, 2003, **11**, 15.
- 7 C. Zhang, X. Wang, J. Lin, B. Ding, J. Yu and N. Pan, *Sens. Actuators, B*, 2011, **152**, 323.
- 8 D. A. Buttry and M. D. Ward, *Chem. Rev.*, 1992, **92**, 1379.
- 9 S. J. Martin, V. E. Granstaff and G. C. Frye, *Anal. Chem.*, 1991, **63**, 2281.
- 10 J. Krim, D. H. Solina and R. Chiarello, *Phys. Rev. Lett.*, 1991, **66**, 184.
- 11 C. J. Percival, S. Stanley, A. Braithwaite, M. I. Newton and G. McHale, *Analyst*, 2002, **127**, 1026.
- 12 C. J. Percival, S. Stanley, M. Galle, A. Braithwaite, M. I. Newton, G. McHale and W. Hayes, *Anal. Chem.*, 2001, **73**, 4228.
- 13 Y. Fu and H. O. Finklea, *Anal. Chem.*, 2003, **75**, 5393.
- 14 N. Andreeva, T. Ishizaki, P. Baroch and N. Saito, *Sens. Actuators, B*, 2012, **164**, 21.
- 15 M. Sun, B. Ding, J. Lin, J. Yu and G. Sun, *Sens. Actuators, B*, 2011, **160**, 434.
- 16 M. Sun, B. Ding and J. Yu, *RSC Adv.*, 2012, **2**, 1378.
- 17 F. Hook, M. Rodahl, P. Brzezinski and B. Kasemo, *Langmuir*, 1998, **14**, 734.
- 18 F. Hook, J. Voros, M. Rodahl, R. Kurrat, P. Boni, J. J. Ramsden, M. Textor, N. D. Spencer, P. Tengvall, J. Gold and B. Kasemo, *Colloids Surf., B*, 2002, **24**, 170.
- 19 K. A. Marx, *Biomacromolecules*, 2003, **4**, 1120.
- 20 B. Ding, H. Kim, C. Kim, M. Khil and S. Park, *Nanotechnology*, 2003, **14**, 537.
- 21 J. Du and Y.-L. Hsieh, *Nanotechnology*, 2008, **19**.
- 22 Z. M. Huang, Y. Z. Zhang, M. Kotaki and S. Ramakrishna, *Compos. Sci. Technol.*, 2003, **63**, 2253.
- 23 D. Li and Y. N. Xia, *Adv. Mater.*, 2004, **16**, 1170.
- 24 A. Greiner and J. H. Wendorff, *Angew. Chem., Int. Ed.*, 2007, **46**, 5703.
- 25 X. Wang, B. Ding and B. Li, *Mater. Today*, 2013, **16**, 241.
- 26 H. Y. Yoo and S. Bruckenstein, *Anal. Chim. Acta*, 2013, **785**, 103.
- 27 X. Wang, B. Ding, G. Sun, M. Wang and J. Yu, *Prog. Mater. Sci.*, 2013, **58**, 1243.
- 28 X. Wang, B. Ding, M. Sun, J. Yu and G. Sun, *Sens. Actuators, B*, 2010, **144**, 17.
- 29 S. Noel, B. Liberelle, L. Robitaille and G. De Crescenzo, *Bioconjugate Chem.*, 2011, **22**, 1699.
- 30 M. E. Kose, B. F. Carroll and K. S. Schanze, *Langmuir*, 2005, **21**, 9129.
- 31 T. Mayr, I. Klimant, O. S. Wolfbeis and T. Werner, *Anal. Chim. Acta*, 2002, **462**, 10.
- 32 Z. Jia and S. Du, *Fibers Polym.*, 2006, **7**, 240.
- 33 G. Z. Sauerbrey, *Physics*, 1959, **155**, 222.
- 34 X. Wang, F. Cui, J. Lin, B. Ding, J. Yu and S. S. Al-Deyab, *Sens. Actuators, B*, 2012, **171–172**, 665.
- 35 S. B. Deng and R. B. Bai, *Water Res.*, 2004, **38**, 2432.
- 36 Y. S. Choi and I. J. Chung, *Polymer*, 2004, **45**, 3834.
- 37 B. Ding, X. Wang, J. Yu and M. Wang, *J. Mater. Chem.*, 2011, **21**, 12792.

- 38 F. Zhou, K. Adachi and K. Kato, *Thin Solid Films*, 2006, **497**, 217.
- 39 X. Wang, J. Wang, Y. Si, B. Ding, J. Yu, G. Sun, W. Luo and G. Zheng, *Nanoscale*, 2012, **4**, 7592.
- 40 C. Lastoskie, K. E. Gubbins and N. Quirke, *Langmuir*, 1993, **9**, 2702.
- 41 P. J. Branton, P. G. Hall, K. S. W. Sing, H. Reichert, F. Schuth and K. K. Unger, *J. Chem. Soc., Faraday Trans.*, 1994, **90**, 2967.
- 42 L. Feng, C. J. Musto and K. S. Suslick, *J. Am. Chem. Soc.*, 2010, **132**, 4047.
- 43 D. Enders, H. Schubert and C. Nubling, *Angew. Chem., Int. Ed.*, 1986, **25**, 1110.
- 44 L. Feng, Y. J. Liu, X. D. Zhou and J. M. Hu, *J. Colloid Interface Sci.*, 2005, **284**, 382.
- 45 J. L. Hu, T. Zhou, Y. F. Zhang, Z. Wang, D. M. Luo and Z. Cao, *Adv. Mater. Res.*, 2011, **233–235**, 723.
- 46 P. B. Shepson, E. Mackay and K. Muthuramu, *Environ. Sci. Technol.*, 1996, **30**, 3623.
- 47 A. Suruda, P. Schulte, M. Boeniger, R. B. Hayes, G. K. Livingston, K. Steenland, P. Stewart, R. Herrick, D. Douthit and M. A. Fingerhut, *Cancer Epidemiol., Biomarkers Prev.*, 1993, **2**, 460.
- 48 A. L. J. Olsson, I. R. Quevedo, D. He, M. Basnet and N. Tufenkji, *ACS Nano*, 2013, **7**, 7843.